Algorithm Sequencing and Curriculum Learning in Deep Reinforcement Learning

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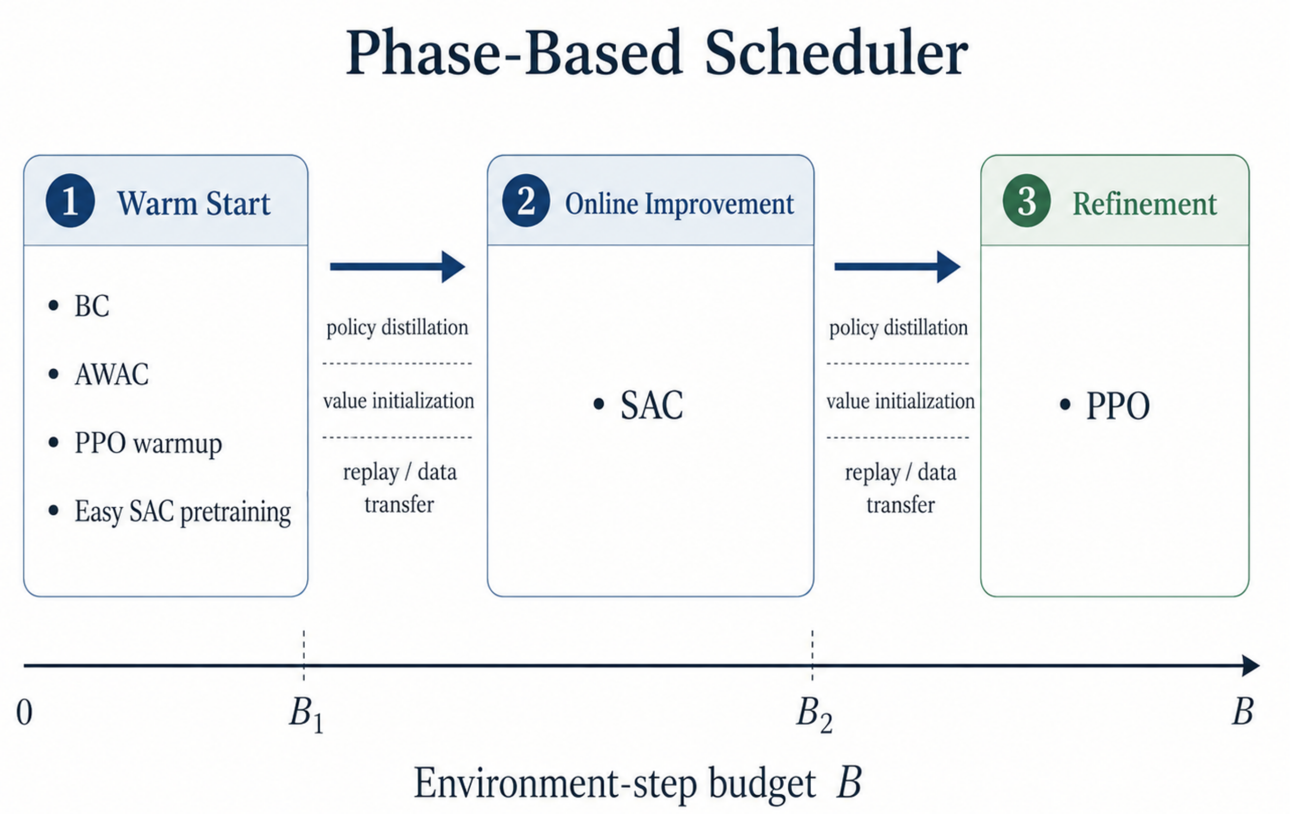
Question and Hypothesis

- RL can run as one fixed algorithm, but training has phases: initialization, exploration, improvement, and refinement
- Can **algorithm sequencing** improve final return, sample efficiency, or stability under matched online budgets?
- Hypothesis: sequencing helps when the first phase gives the next phase a useful policy, state distribution, or representation while leaving enough budget to improve
- **Phase compatibility matters and needs to be understood**



Experiment Map

Family	Question	Evidence
Single algorithms	How strong are fixed baselines?	PPO, SAC; Hopper/Walker.
Online handoffs	Does switch direction/timing matter?	SAC→PPO, PPO→SAC; 25/50/75% and plateau switches.
Transfer mechanism	Does value warmup fix SAC→PPO?	Random, self-warmup, source-aligned handoff PPO values.
Offline starts	Can BC/AWAC initialize online RL?	BC→SAC, AWAC→SAC, interleaved BC anchoring, PPO targets.
Curriculum starts	Does an easier first phase help?	Easy SAC→SAC on Hopper; "easy" ignores early termination during pretraining.

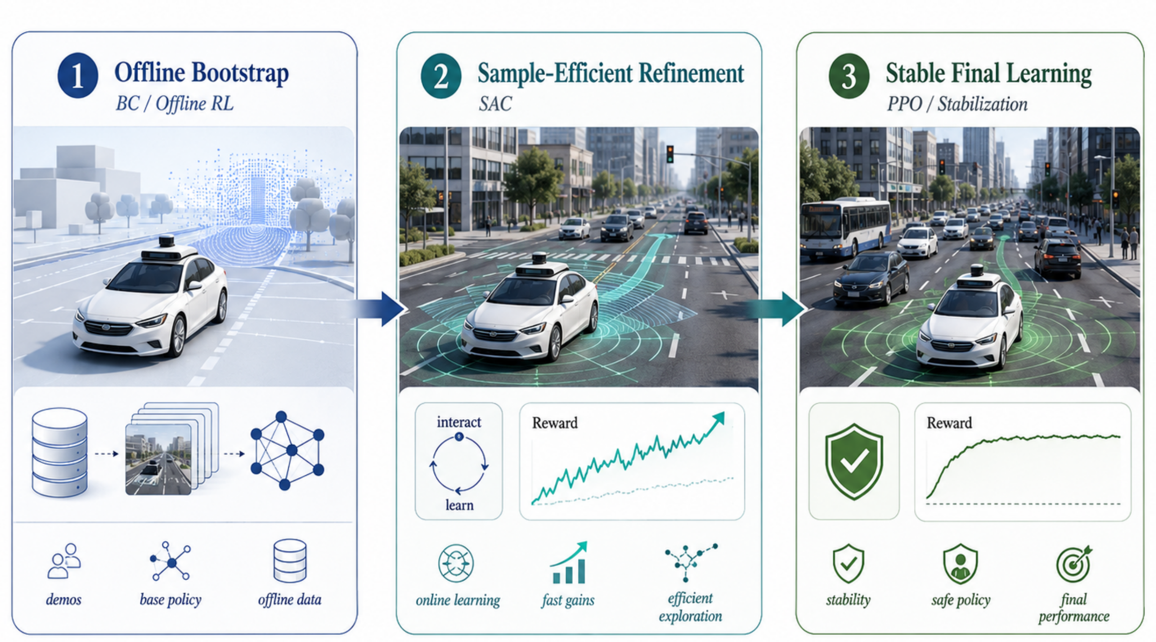
What We Measure

- **Final return:** end-of-budget policy quality
- **Normalized AUC:** sample efficiency over the full learning curve
- **Stability:** seed variance, collapse count, and worst-seed return
- **Budget accounting:** online environment steps are matched. Offline data and updates are reported separately

Design Principle

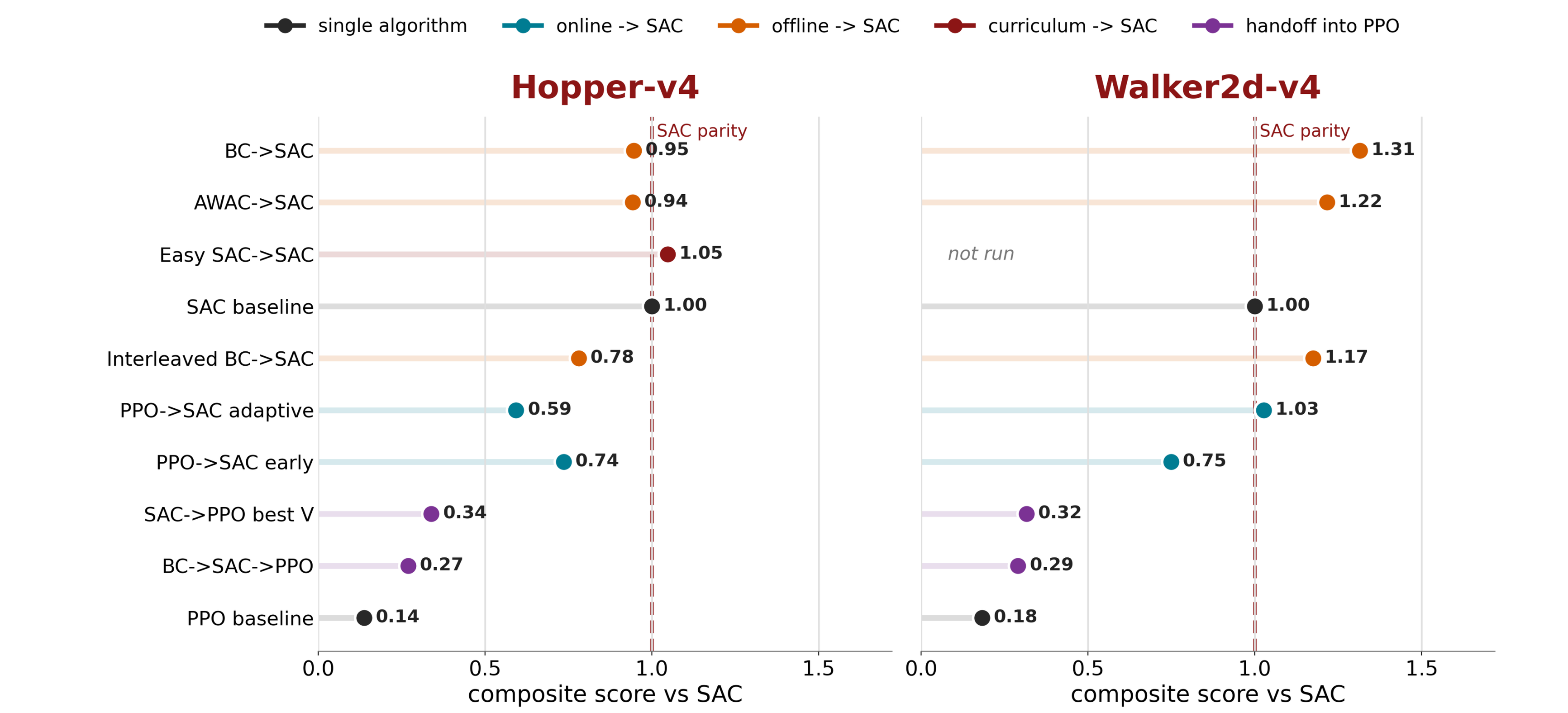
- Warm starts should produce behavior the online learner can preserve and improve
- Exploration phases should hand off early enough for the target algorithm to use most of the remaining budget
- Refinement phases need transfer mechanisms that prevent policy degradation. Value initialization alone is weak evidence for this

Motivation: Three-Stage Policy Training



Main Result: Compatible Schedules End in SAC

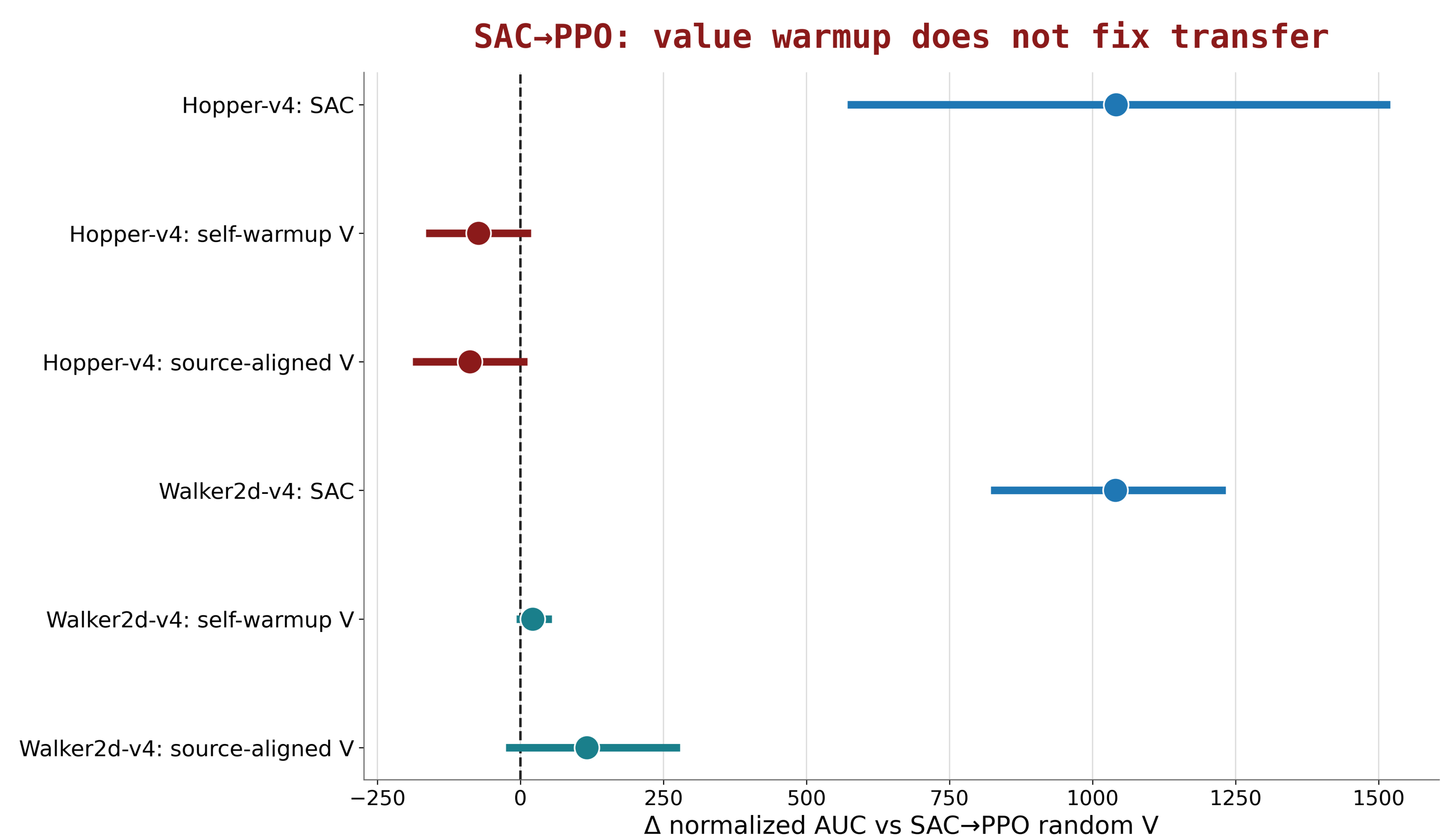
Synthesis: schedules that end in SAC dominate the phase-allocation frontier



Composite = 0.5 x final-return ratio + 0.5 x AUC ratio relative to the matched SAC baseline in that environment.

- SAC remains the strongest fixed baseline, so a useful scheduler must justify its phase allocation against SAC
- The best non-baseline schedules hand into SAC: PPO→SAC, BC→SAC, AWAC→SAC, interleaved BC→SAC, and Easy SAC→SAC
- Handoffs into PPO preserve neither final return nor AUC, even when they receive a pretrained policy

Transfer Out of SAC Is Fragile



95% paired CIs; zero indicates no gain over random value initialization.

- SAC→PPO did not become reliable after PPO value warmup compared to SAC only baseline
- There is a mismatch between source trajectory and target learner preventing all handoffs from learning
- PPO refinement needs a **stronger policy-retention** or **state-distribution transfer mechanism** before it can be a dependable final phase

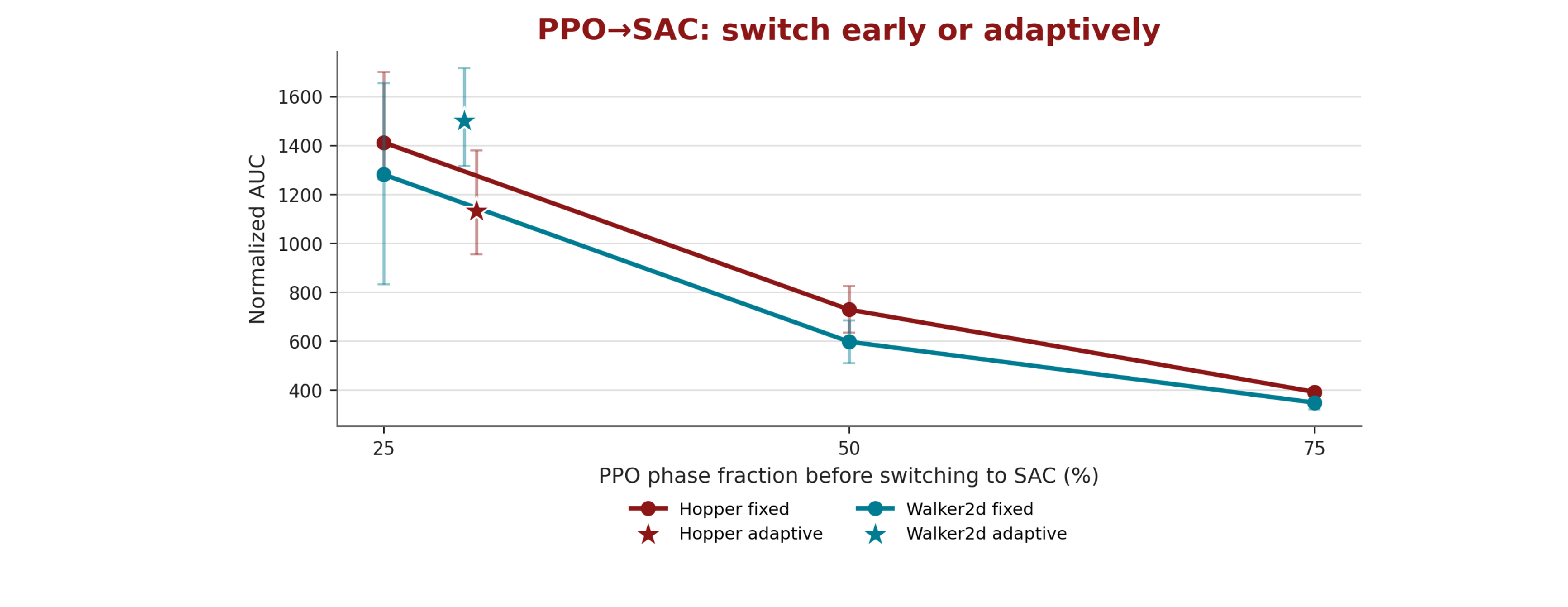
Curriculum / Transfer Learning

Easy Hopper: We modify Hopper-v4 only during pretraining by ignoring early failure terminations. If the hopper falls, the episode continues, giving SAC longer trajectories before transferring the learned policy to normal Hopper-v4

- Tests whether a forgiving first phase can learn reusable locomotion behavior before the agent faces the real termination conditions
- Does this produces a better initialization for real environment SAC fine-tuning?

PPO→SAC Needs Early or Adaptive Switching

Adaptive rule: start with PPO, require a 25% minimum phase, then switch to SAC after 3 evaluation checkpoints whose mean return does not exceed the best return seen so far

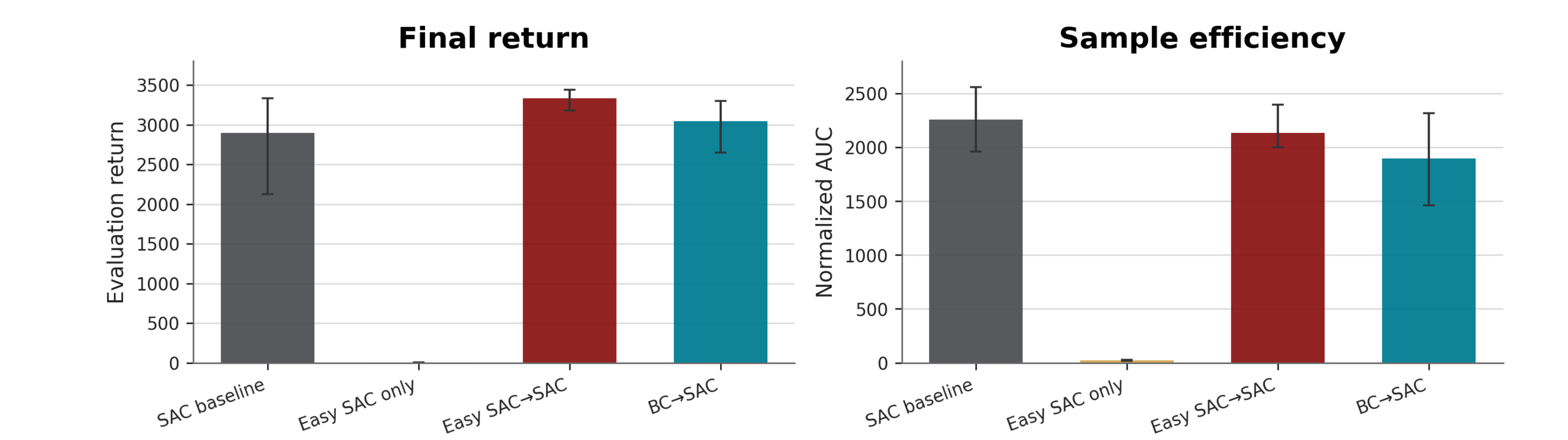


Stars show plateau-triggered adaptive switches; both fire near the strong early-switch regime.

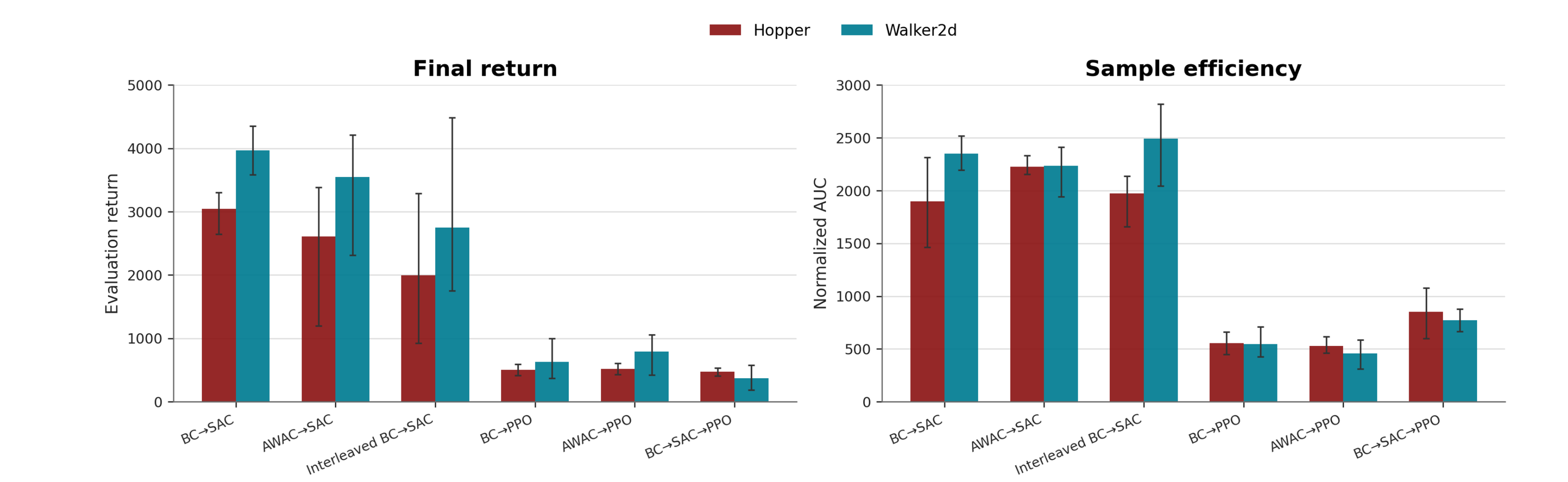
- Fixed 25% PPO phases beat late 75% switches; SAC needs enough remaining budget to improve the distilled policy
- Plateau switches fire near the strong early-switch regime and avoid wasting budget on an exhausted first phase

Curriculum and Offline Starts Work Best Into SAC

Curriculum transfer: forgiving SAC pretrain → real SAC



Hopper-v4, 500k online steps. "Easy" pretraining ignores MuJoCo termination, then distills into SAC on the real task.



Completed offline sources: BC, AWAC, and interleaved BC anchoring. 500k online steps; error bars show bootstrap 95% CIs.

- Easy SAC→SAC tests curriculum learning: pretrain in a forgiving Hopper variant, then distill into SAC on real task
- BC/AWAC/interleaved starts transfer best into SAC. PPO targets remain weak after the same starts

Takeaways and Next Steps

- **Transitions into SAC** are more effective than transitions out of SAC
- **Early/adaptive handoffs** outperform late handoffs
- Ending in SAC allows replay-based off-policy updates to exploit warm starts, and PPO can degrade transferred behavior
- BC/AWAC starts help only when the next online learner preserves and improves the policy
- Develop more robust adaptive switching rules using richer signals such as entropy, critic stability, KL drift, and return plateau rather than return plateau alone

[1] Narvekar et al., *Curriculum Learning for Reinforcement Learning Domains*, JMLR 2020.
[2] Taylor and Stone, *Transfer Learning for Reinforcement Learning Domains: A Survey*, JMLR 2009.